

The (λ, N) sweep on QED marginal against edlm, MDLM and UDLM

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1 Setup

196 runs, 1,024 samples each at $T = 32$ steps, seed 1, reward `qed`. Grid: $\lambda \in \{0, 2, 5, 20, 50, 200, 1000\}$ against $N \in \{10, 30, 100, 300, 500, 1000, 2000\}$, for both mixture-sampling modes and both base models.

N is the number of $\mathbf{x}_0$ candidates drawn per denoising step — the Monte Carlo budget of the estimator. λ is the tilt in $q(\mathbf{x}_0) \propto p_\theta(\mathbf{x}_0) e^{\lambda r(\mathbf{x}_0)}$. The two modes differ only in how $\mathbf{x}_{t-1}$ is drawn from the resulting mixture: `marginal` averages the N individually normalised posteriors, `edlm` draws a component and then samples from it. They share per-position marginals and differ only in the joint.

2 Reading ESS: use the median, never the last step

Effective sample size $\text{ESS} = 1 / \sum_n w_n^2$ says which end of the grid a run is on: $\text{ESS} = N$ means the tilt is doing nothing, $\text{ESS} \rightarrow 1$ means it has collapsed onto a single candidate.

The driver script prints only the *final* step’s ESS, and that number is actively misleading on MDLM. Absorbing-state sampling has almost everything unmasked by the last step, so all N candidates decode to the same sequence and tie on reward — the last-step ESS reads $\approx N$ however hard λ is tilting. Judged on it, MDLM looks entirely unguided. Its novel QED in fact climbs across the λ grid. Every ESS below is the **median over the trajectory**, digested from the per-run logs into `results/_ess_summary.csv`.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	1.00	0.97	0.87	0.81	0.77	0.74	0.73
30	1.00	0.96	0.83	0.72	0.67	0.63	0.62
100	1.00	0.96	0.81	0.66	0.60	0.54	0.54
300	1.00	0.96	0.78	0.57	0.50	0.44	0.43
500	1.00	0.96	0.78	0.56	0.49	0.43	0.41
1000	0.00	0.95	0.77	0.54	0.45	0.40	0.40
2000	0.00	0.95	0.73	0.43	0.40	0.34	0.32

Table 1: MDLM, `marginal`: median ESS as a fraction of N .

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	1.00	0.97	0.89	0.82	0.78	0.74	0.73
30	1.00	0.96	0.84	0.72	0.67	0.63	0.62
100	1.00	0.96	0.80	0.62	0.56	0.51	0.50
300	1.00	0.96	0.77	0.57	0.49	0.44	0.44
500	1.00	0.96	0.78	0.55	0.48	0.43	0.41
1000	0.00	0.95	0.76	0.53	0.46	0.41	0.39
2000	0.00	0.95	0.73	0.47	0.39	0.34	0.32

Table 2: MDLM, `edlm`: median ESS as a fraction of N .

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	1.00	0.96	0.85	0.78	0.76	0.76	0.76
30	1.00	0.95	0.77	0.60	0.56	0.56	0.56
100	1.00	0.95	0.65	0.43	0.37	0.34	0.34
300	1.00	0.95	0.57	0.22	0.19	0.17	0.16
500	1.00	0.94	0.54	0.17	0.15	0.13	0.12
1000	0.00	0.95	0.51	0.10	0.09	0.08	0.08
2000	0.00	0.94	0.49	0.07	0.06	0.05	0.05

Table 3: UDLM, `marginal`: median ESS as a fraction of N .

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	1.00	0.96	0.86	0.79	0.77	0.76	0.76
30	1.00	0.95	0.76	0.60	0.56	0.56	0.56
100	1.00	0.95	0.67	0.42	0.37	0.35	0.35
300	1.00	0.94	0.56	0.22	0.20	0.18	0.17
500	1.00	0.95	0.53	0.16	0.14	0.13	0.12
1000	0.00	0.95	0.52	0.11	0.10	0.08	0.08
2000	0.00	0.94	0.49	0.07	0.06	0.05	0.05

Table 4: UDLM, `edlm`: median ESS as a fraction of N .

3 The grid

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	0.456	0.456	0.460	0.467	0.478	0.480	0.482
30	0.457	0.459	0.455	0.470	0.487	0.492	0.491
100	0.464	0.459	0.460	0.478	0.493	0.495	0.499
300	0.455	0.456	0.459	0.478	0.497	0.509	0.512
500	0.456	0.457	0.462	0.483	0.500	0.511	0.515
1000	0.455	0.459	0.460	0.479	0.501	0.515	0.516
2000	0.459	0.465	0.464	0.486	0.503	0.519	0.516

Table 5: MDLM, `marginal`: novel-QED mean.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	0.458	0.465	0.468	0.473	0.474	0.479	0.480
30	0.451	0.455	0.459	0.478	0.487	0.489	0.492
100	0.458	0.469	0.460	0.483	0.495	0.501	0.503
300	0.452	0.455	0.459	0.481	0.495	0.501	0.504
500	0.454	0.453	0.467	0.484	0.503	0.510	0.512
1000	0.459	0.461	0.467	0.484	0.498	0.513	0.514
2000	0.454	0.460	0.461	0.486	0.506	0.520	0.518

Table 6: MDLM, `edlm`: novel-QED mean.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	0.452	0.452	0.451	0.476	0.492	0.506	0.502
30	0.457	0.460	0.458	0.487	0.505	0.515	0.522
100	0.456	0.464	0.464	0.489	0.506	0.521	0.529
300	0.459	0.459	0.459	0.485	0.513	0.533	0.533
500	0.452	0.463	0.462	0.484	0.514	0.536	0.540
1000	0.454	0.454	0.468	0.484	0.516	0.542	0.551
2000	0.463	0.467	0.463	0.492	0.511	0.547	0.548

Table 7: UDLM, `marginal`: novel-QED mean.

4 marginal against edlm

Two things were predicted before the sweep ran, and only one holds.

Predicted and confirmed: the two modes are empirically interchangeable. Across the whole grid the mean absolute gap in novel-QED is 0.006 with mean signed gap -0.003 — no consistent winner, at any λ or N .

Predicted and *not* confirmed: the gap was expected to *shrink* as λ grows, because one-hot weights make $\sum_n w_n q(\cdot | \mathbf{x}_0^{(n)})$ collapse to the single component `edlm` would have drawn, and a posterior conditioned on a fixed $\mathbf{x}_0$ already factorises over positions. Measured, the gap is flat: 0.005 at $\lambda = 0$ against 0.005 at $\lambda = 1000$. The reason is visible in the ESS tables — the weights

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	0.446	0.454	0.462	0.482	0.501	0.512	0.510
30	0.459	0.466	0.475	0.488	0.508	0.521	0.525
100	0.457	0.468	0.468	0.502	0.519	0.533	0.535
300	0.449	0.462	0.468	0.494	0.530	0.543	0.542
500	0.441	0.450	0.462	0.503	0.525	0.544	0.549
1000	0.451	0.456	0.470	0.497	0.531	0.544	0.546
2000	0.455	0.465	0.471	0.502	0.534	0.552	0.556

Table 8: UDLM, **edlm**: novel-QED mean.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	216	222	246	253	262	263	268
30	211	216	247	275	283	297	295
100	211	214	236	287	286	305	314
300	220	229	234	313	319	381	389
500	228	244	270	327	348	386	391
1000	204	222	252	344	367	424	410
2000	190	206	234	342	386	410	439

Table 9: MDLM, **marginal**: number of novel molecules (out of 1,024 generated).

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	215	231	234	244	247	248	248
30	234	236	256	270	277	297	302
100	208	231	260	305	314	332	336
300	209	238	238	301	330	356	358
500	195	198	233	295	345	351	369
1000	219	234	252	349	373	415	433
2000	213	227	238	343	370	426	434

Table 10: MDLM, **edlm**: number of novel molecules (out of 1,024 generated).

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	189	211	229	239	236	232	231
30	195	201	205	246	238	245	269
100	194	233	241	288	259	279	288
300	193	214	217	251	259	316	311
500	196	225	229	275	274	334	333
1000	186	207	236	271	274	345	371
2000	207	236	242	257	295	359	392

Table 11: UDLM, **marginal**: number of novel molecules (out of 1,024 generated).

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	165	212	238	222	228	240	239
30	191	224	240	252	236	254	272
100	180	214	230	300	263	307	297
300	179	200	251	327	290	306	326
500	190	210	247	266	274	306	355
1000	164	207	241	256	310	331	366
2000	212	215	263	300	297	374	397

Table 12: UDLM, **edlm**: number of novel molecules (out of 1,024 generated).

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	0.636	0.636	0.636	0.644	0.674	0.674	0.684
30	0.696	0.696	0.673	0.692	0.692	0.692	0.692
100	0.642	0.642	0.642	0.695	0.695	0.682	0.695
300	0.667	0.673	0.673	0.635	0.680	0.688	0.688
500	0.629	0.629	0.668	0.659	0.662	0.692	0.697
1000	0.644	0.644	0.687	0.649	0.705	0.697	0.697
2000	0.628	0.628	0.641	0.670	0.682	0.688	0.688

A raw max grows with the number of molecules it is taken over, and these cells differ several-fold in novel count, so a higher max here is partly a deeper pool. That is a real advantage at a fixed generation budget – every cell generated the same 1,024 sequences – but it is not a claim about distribution shape. Section 7 separates the two.

Table 13: MDLM, **marginal**: highest novel-QED of the run.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	0.644	0.644	0.680	0.638	0.644	0.644	0.644
30	0.628	0.628	0.630	0.648	0.683	0.690	0.690
100	0.627	0.625	0.622	0.646	0.667	0.670	0.670
300	0.624	0.660	0.660	0.684	0.657	0.694	0.684
500	0.645	0.645	0.643	0.650	0.684	0.688	0.688
1000	0.614	0.636	0.652	0.658	0.670	0.696	0.687
2000	0.643	0.652	0.644	0.688	0.666	0.699	0.690

Table 14: MDLM, **edlm**: highest novel-QED of the run.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	0.622	0.623	0.624	0.643	0.645	0.658	0.656
30	0.680	0.659	0.631	0.693	0.693	0.661	0.686
100	0.616	0.640	0.683	0.655	0.673	0.688	0.697
300	0.601	0.680	0.680	0.651	0.654	0.701	0.701
500	0.611	0.648	0.647	0.697	0.693	0.690	0.687
1000	0.637	0.632	0.638	0.685	0.730	0.730	0.730
2000	0.675	0.644	0.620	0.663	0.662	0.710	0.733

Table 15: UDLM, **marginal**: highest novel-QED of the run.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	0.615	0.624	0.653	0.675	0.675	0.660	0.660
30	0.636	0.652	0.655	0.693	0.665	0.666	0.675
100	0.648	0.661	0.623	0.670	0.651	0.675	0.672
300	0.617	0.629	0.655	0.657	0.661	0.716	0.677
500	0.647	0.645	0.656	0.666	0.663	0.700	0.735
1000	0.619	0.657	0.655	0.705	0.713	0.709	0.714
2000	0.663	0.663	0.635	0.693	0.677	0.707	0.711

Table 16: UDLM, `edlm`: highest novel-QED of the run.

model	λ	mean $ \Delta $ property	mean Δ property	mean $ \Delta $ novel	mean Δ novel
MDLM	0	0.004	+0.002	15.6	-1.9
MDLM	2	0.005	-0.001	19.1	-6.0
MDLM	5	0.004	-0.003	12.9	+1.1
MDLM	20	0.004	-0.004	11.7	+4.9
MDLM	50	0.002	+0.000	12.1	-0.7
MDLM	200	0.003	+0.001	18.1	+5.9
MDLM	1000	0.003	+0.001	18.6	+3.7
UDLM	0	0.006	+0.005	12.7	+11.3
UDLM	2	0.005	-0.001	13.3	+6.4
UDLM	5	0.007	-0.007	19.0	-15.9
UDLM	20	0.010	-0.010	25.4	-13.7
UDLM	50	0.013	-0.013	11.9	-9.0
UDLM	200	0.007	-0.007	16.0	-1.1
UDLM	1000	0.007	-0.005	9.6	-8.1

A systematic winner would show as a consistent sign in the Δ columns. There is none.

Table 17: `marginal` minus `edlm`, averaged over the five N at each λ . $|\Delta|$ is the typical size of the gap, Δ its sign.

never actually become one-hot in this grid, so the limit is never entered. The flat gap is therefore sampling noise between two samplers that agree, which is the same conclusion by a weaker route.

5 $\lambda = 0$: the unguided control

At $\lambda = 0$ the weights are uniform, so no tilt is applied whatever N or the mode is. Every cell of Table 18 should therefore be the base sampler, and the spread across it measures the sampling noise that every other comparison in this report has to clear.

model	mode	$N = 10$	$N = 30$	$N = 100$	$N = 300$	$N = 500$	$N = 1000$	$N = 2000$
MDLM	marginal	0.456	0.457	0.464	0.455	0.456	0.455	0.459
MDLM	edlm	0.458	0.451	0.458	0.452	0.454	0.459	0.454
UDLM	marginal	0.452	0.457	0.456	0.459	0.452	0.454	0.463
UDLM	edlm	0.446	0.459	0.457	0.449	0.441	0.451	0.455

This is the implementation control: at $\lambda = 0$ the weights are uniform, so no tilt is applied and the numbers must be flat in N up to sampling noise. They are.

Table 18: novel-QED at $\lambda = 0$. Every entry should be the base sampler, independent of N and of the mode.

This table also calibrates everything else in the report. The widest $\lambda = 0$ spread within a single model is 0.021 in novel-QED — runs that are identical in distribution, differing only by sampling noise. The mean gap between **marginal** and **edlm** across the whole grid is 0.006, **smaller** than that floor. So the two modes are not merely close: their difference is indistinguishable from noise.

6 The frontier against D-CBG

Both methods are swept, then compared on the axes the paper’s Figure 3 uses: number of novel molecules against novel-QED mean. Comparing at single hyperparameter points would be meaningless — the paper tunes γ per model *and* per property.

method	setting	valid	novel	novel-QED
ours	edlm, $N = 2000$, $\lambda = 200$	747	426	0.520
ours	edlm, $N = 2000$, $\lambda = 1000$	747	434	0.518
ours	marginal, $N = 2000$, $\lambda = 1000$	750	439	0.516
D-CBG	$\gamma = 3$, exact	390	100	0.588
D-CBG	$\gamma = 2$, exact	488	127	0.574
D-CBG	$\gamma = 1$, exact	537	189	0.557
D-CBG	$\gamma = 1$, approx.	504	208	0.467

Table 19: MDLM: the Pareto-optimal settings of each method on (novel count, novel-QED).

7 Beyond the mean

Everything above compares *means*, which is the paper’s metric but not the one a screening pipeline faces. Nobody consumes the mean of a generated library; they take the usable molecules out of it.

method	setting	valid	novel	novel-QED
ours	edlm, $N = 2000$, $\lambda = 1000$	923	397	0.556
D-CBG	$\gamma = 10$, exact	989	63	0.609
D-CBG	$\gamma = 5$, exact	968	82	0.593
D-CBG	$\gamma = 3$, exact	946	102	0.588
D-CBG	$\gamma = 2$, exact	903	115	0.567
D-CBG	$\gamma = 1$, exact	949	118	0.528
D-CBG	$\gamma = 1$, approx.	997	121	0.477

Table 20: UDLM: the Pareto-optimal settings of each method on (novel count, novel-QED).

And the mean is blind to the axis these methods differ on most: from the same 1,024 generated sequences our runs yield several times more novel molecules than D-CBG’s.

7.1 The primary comparison: usable molecules per fixed budget

Both methods generated exactly 1,024 sequences. So simply count the novel molecules that clear a quality bar. Quality and quantity both enter — a run that yields many poor novel molecules scores nothing, and one that yields a handful of excellent ones scores little.

This number needs no correction of any kind. In particular it must **not** be normalised by the novel count, nor size-matched against the competitor: converting a fixed budget into more usable molecules *is* the result, and dividing it out would erase precisely what is being measured.

7.1.1 The whole grid, on the primary metric

The tables above pick a best setting per method, which is what a headline needs but not what a reader should have to trust. Below is the raw grid: hits at the headline bar at every (N, λ) , both modes, nothing averaged and nothing filtered. It shows where in the grid the hits live — and that the surface is not monotone in either axis, so a single “best” cell is partly a draw from run noise.

N	marg. 0	marg. 20	marg. 50	marg. 200	marg. 1000	edlm. 0	edlm. 20	edlm. 50	edlm. 200	edlm. 1000
10	2	7	11	12	13	5	9	10	10	11
30	9	8	12	19	18	3	14	15	17	18
100	6	10	20	22	29	6	13	19	24	31
300	3	12	23	39	39	2	9	23	27	33
500	4	18	19	37	41	6	14	33	38	38
1000	5	11	23	45	48	3	12	26	40	47
2000	3	19	25	56	53	3	10	35	63	50

Novel molecules clearing the bar, out of the 1,024 generated. Every cell is a single run, seed 1; nothing is averaged. $\lambda \in \{2, 5\}$ are omitted to fit the page. **Not** filtered by `MIN_NOVEL`, unlike the Pareto table — a collapsed run simply scores few hits here, which is the honest reading, whereas a mean over 2 molecules is not. **D-CBG’s best on this bar is 40** (exact $\gamma = 2$), for comparison.

Table 21: MDLM: hits at $\text{QED} \geq 0.6$ across the whole (N, λ) grid. Columns are (mixture mode) $\times \lambda$.

N	marg. 0	marg. 20	marg. 50	marg. 200	marg. 1000	edlm. 0	edlm. 20	edlm. 50	edlm. 200	edlm. 1000
10	7	9	13	15	11	2	14	15	24	23
30	5	12	27	17	27	8	18	18	17	27
100	7	17	20	35	31	4	15	31	38	42
300	1	9	21	43	47	4	8	34	55	59
500	2	13	28	54	66	7	16	38	72	87
1000	3	11	33	60	83	1	14	51	67	86
2000	9	21	29	82	93	5	25	56	91	111

Novel molecules clearing the bar, out of the 1,024 generated. Every cell is a single run, seed 1; nothing is averaged. $\lambda \in \{2, 5\}$ are omitted to fit the page. **Not** filtered by `MIN_NOVEL`, unlike the Pareto table — a collapsed run simply scores few hits here, which is the honest reading, whereas a mean over 2 molecules is not. **D-CBG’s best on this bar is 46** (exact $\gamma = 3$), for comparison.

Table 22: UDLM: hits at QED ≥ 0.6 across the whole (N, λ) grid. Columns are (mixture mode) $\times \lambda$.

model	method	setting	novel	mean	≥ 0.55	≥ 0.6	≥ 0.65
MDLM	marginal	$N = 2000, \lambda = 200$	410	0.519	129 (1.3 $\times$)	56 (1.4 $\times$)	8 (0.9 $\times$)
MDLM	edlm	$N = 2000, \lambda = 200$	426	0.520	130 (1.4 $\times$)	63 (1.6 $\times$)	8 (0.9 $\times$)
MDLM	D-CBG	$\gamma = 2$, exact	127	0.574	96	40	9
UDLM	marginal	$N = 2000, \lambda = 1000$	392	0.548	198 (2.1 $\times$)	93 (2.0 $\times$)	20 (5.0 $\times$)
UDLM	edlm	$N = 2000, \lambda = 1000$	397	0.556	223 (2.4 $\times$)	111 (2.4 $\times$)	21 (5.2 $\times$)
UDLM	D-CBG	$\gamma = 3$, exact	102	0.588	93	46	4

No size matching and no normalisation — every run spent the same generation budget, so these are directly comparable counts.

Table 23: Novel molecules clearing each QED bar, out of the 1,024 sequences each run generated. The two mixture-sampling modes are kept apart rather than merged into a single *ours* row, since they are different samplers. Each is at its own best setting, ranked on the middle bar; the multiplier is against D-CBG on the same model.

7.2 Secondary: shape comparisons, which discard the counts

Two standard distribution-free comparisons. Both deliberately ignore how many molecules each method produced, so neither can see the advantage above; they are here to answer a different question — whether our *distribution* is better, or only our yield.

- **Rank against rank.** Sort both sets and compare like for like: best against best, tenth percentile against tenth. The fraction of quantiles where we lead is the *quantile win rate*; 1.0 would be **first-order stochastic dominance**. Comparing the k -th *rank* directly would need equal sample sizes, so this compares at matched quantiles, the size-independent form of the same idea.
- **Draw against draw.** A_{12} , the Vargha–Delaney probability of superiority: one molecule drawn at random from each, how often is ours better? The normalised Mann–Whitney U ; 0.5 is indistinguishable.

The *matched* column subsamples the larger novel set down to the smaller. It is a **decomposition, not a fairness correction** — it separates “we win because our tail is better” from “we win because our pool is deeper”. The counterfactual it describes never occurs in practice, so it should not be read as the fair comparison; the table above is.

model	method	novel	mean	max	top10	top10 matched	q-win	A_{12}
MDLM	ours	391	0.515	0.697	0.679	0.639	0.00	0.20
MDLM	D-CBG	100	0.588	0.697	0.672	0.672	–	–
UDLM	ours	392	0.548	0.733	0.693	0.646	0.05	0.21
UDLM	D-CBG	63	0.609	0.656	0.649	0.649	–	–

max and *top-K* are at the fixed 1,024-sequence budget and so include our deeper novel pool. *matched* removes that, and answers only “quality or quantity?”.

Table 24: Each method at its own best setting by top-10. *q-win* and A_{12} sit on the ours row and describe ours against D-CBG.

8 Validity

Validity is the axis on which the two mechanisms differ most sharply, and it is what drives the curve directions of the previous section, so it is worth reporting on its own.

Our reward sees a *clean* x_0 candidate and returns `invalid_reward` = 0 when RDKit cannot parse it. Raising λ therefore pushes probability towards parseable molecules before it pushes towards high QED: an invalid candidate is treated as a QED-zero one and suppressed. D-CBG instead perturbs the per-position logits with the gradient of a noisy classifier, which carries no such term — nothing in it prefers a parseable string.

9 The shape of each curve

Plot novel count on the x -axis against novel-QED mean on the y -axis and sweep each method’s strength dial — λ for ours, γ for D-CBG. The two methods trace curves in *opposite directions*, and

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	547	571	601	638	646	647	648
30	542	555	596	650	650	655	659
100	532	545	593	662	665	666	663
300	552	577	609	699	706	713	720
500	558	581	610	708	719	718	718
1000	549	578	605	718	716	724	715
2000	544	568	593	736	749	741	750

Table 25: MDLM, **marginal**: valid molecules out of the 1,024 generated.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	564	589	605	634	631	630	630
30	558	583	623	670	670	673	675
100	553	588	624	725	727	721	716
300	556	591	630	715	717	719	713
500	549	570	614	728	747	724	722
1000	542	573	609	738	744	736	739
2000	559	601	622	753	758	747	747

Table 26: MDLM, **edlm**: valid molecules out of the 1,024 generated.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	682	730	801	893	885	886	892
30	695	764	810	920	924	920	918
100	686	758	834	934	921	910	929
300	672	742	844	929	917	925	921
500	691	761	822	934	920	910	916
1000	695	757	835	924	911	927	901
2000	694	770	830	937	926	911	903

Table 27: UDLM, **marginal**: valid molecules out of the 1,024 generated.

N	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
10	657	746	842	900	894	890	891
30	675	761	874	936	929	930	928
100	679	744	845	952	932	916	921
300	693	772	860	963	957	948	938
500	686	785	863	966	948	928	927
1000	685	763	856	964	954	923	910
2000	685	759	863	969	964	927	923

Table 28: UDLM, **edlm**: valid molecules out of the 1,024 generated.

model	method	weakest dial	best	strongest dial	trend
MDLM	ours ($N = 2000$)	544	750	750	rises
MDLM	D-CBG exact	537	537	33	falls
MDLM	D-CBG approx.	504	504	108	falls
UDLM	ours ($N = 2000$)	694	937	903	rises
UDLM	D-CBG exact	949	989	989	rises
UDLM	D-CBG approx.	997	997	819	falls

This is the mechanism behind the opposite curve directions: our dial buys validity on the way to buying property, so the novel count rises with it, while D-CBG spends validity and the novel count falls. On MDLM the D-CBG exact arm reaches zero valid molecules at $\gamma \geq 6$, which is why the paper does not report it there.

Table 29: Valid molecules out of 1,024 as the strength dial is turned up. Ours is read along λ at the largest N ; D-CBG along γ .

that is a property worth reporting on its own, independently of which sits higher.

- **D-CBG runs up and to the left.** Raising γ buys property by spending novelty: the dial is a genuine trade-off the user has to tune, and the original paper tunes it per model and per property.
- **Ours runs up and to the right.** Raising λ improves *both* axes at once. The reason is visible in the validity column: the first thing λ buys is valid molecules, which raises the novel count, and only then does it buy property. There is no trade-off to tune — the only cost is compute.

The correlation between the two axes along each sweep reduces this to one number. Crucially it is measured at *fixed* N , so the up-right shape is not an artifact of spending more Monte Carlo budget: along every row below, only the strength dial moves.

model	method	pts	weakest dial	strongest dial	corr	shape
MDLM	ours, $N = 100$	7	(211, 0.464)	(314, 0.499)	+0.94	up-right
MDLM	ours, $N = 300$	7	(220, 0.455)	(389, 0.512)	+0.98	up-right
MDLM	ours, $N = 1000$	7	(204, 0.455)	(410, 0.516)	+0.97	up-right
MDLM	ours, $N = 2000$	7	(190, 0.459)	(439, 0.516)	+0.98	up-right
MDLM	D-CBG exact	5	(189, 0.557)	(14, 0.575)	-0.43	up-left
MDLM	D-CBG approx.	5	(208, 0.467)	(45, 0.556)	-0.84	up-left
UDLM	ours, $N = 100$	7	(194, 0.456)	(288, 0.529)	+0.83	up-right
UDLM	ours, $N = 300$	7	(193, 0.459)	(311, 0.533)	+0.96	up-right
UDLM	ours, $N = 1000$	7	(186, 0.454)	(371, 0.551)	+0.97	up-right
UDLM	ours, $N = 2000$	7	(207, 0.463)	(392, 0.548)	+0.97	up-right
UDLM	D-CBG exact	5	(118, 0.528)	(63, 0.609)	-0.86	up-left
UDLM	D-CBG approx.	5	(121, 0.477)	(59, 0.556)	-0.98	up-left

Measured at fixed N , so the up-right shape is not an effect of raising the Monte Carlo budget — only the strength dial moves along each row.

Table 30: Direction of each curve in (novel count, novel-QED) space. *corr* is between the two axes along the sweep: positive means the dial improves both, negative means it trades one for the other.

ours	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
$N = 10$	(216, 0.456)	(222, 0.456)	(246, 0.460)	(253, 0.467)	(262, 0.478)	(263, 0.480)	(268, 0.482)
$N = 30$	(211, 0.457)	(216, 0.459)	(247, 0.455)	(275, 0.470)	(283, 0.487)	(297, 0.492)	(295, 0.491)
$N = 100$	(211, 0.464)	(214, 0.459)	(236, 0.460)	(287, 0.478)	(286, 0.493)	(305, 0.495)	(314, 0.499)
$N = 300$	(220, 0.455)	(229, 0.456)	(234, 0.459)	(313, 0.478)	(319, 0.497)	(381, 0.509)	(389, 0.512)
$N = 500$	(228, 0.456)	(244, 0.457)	(270, 0.462)	(327, 0.483)	(348, 0.500)	(386, 0.511)	(391, 0.515)
$N = 1000$	(204, 0.455)	(222, 0.459)	(252, 0.460)	(344, 0.479)	(367, 0.501)	(424, 0.515)	(410, 0.516)
$N = 2000$	(190, 0.459)	(206, 0.465)	(234, 0.464)	(342, 0.486)	(386, 0.503)	(410, 0.519)	(439, 0.516)

Table 31: MDLM: the (novel, novel-QED) point at each λ , for every N . Reading a row left to right traces that curve.

D-CBG						
exact	$\gamma = 1$: (189, 0.557)	$\gamma = 2$: (127, 0.574)	$\gamma = 3$: (100, 0.588)	$\gamma = 4$: (56, 0.566)	$\gamma = 5$: (14, 0.575)	
approx.	$\gamma = 1$: (208, 0.467)	$\gamma = 2$: (165, 0.464)	$\gamma = 3$: (145, 0.472)	$\gamma = 5$: (80, 0.488)	$\gamma = 10$: (45, 0.556)	

Table 32: MDLM: the same for D-CBG, sweeping γ . The novel count falls as the property rises — the opposite direction.

ours	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 20$	$\lambda = 50$	$\lambda = 200$	$\lambda = 1000$
$N = 10$	(189, 0.452)	(211, 0.452)	(229, 0.451)	(239, 0.476)	(236, 0.492)	(232, 0.506)	(231, 0.502)
$N = 30$	(195, 0.457)	(201, 0.460)	(205, 0.458)	(246, 0.487)	(238, 0.505)	(245, 0.515)	(269, 0.522)
$N = 100$	(194, 0.456)	(233, 0.464)	(241, 0.464)	(288, 0.489)	(259, 0.506)	(279, 0.521)	(288, 0.529)
$N = 300$	(193, 0.459)	(214, 0.459)	(217, 0.459)	(251, 0.485)	(259, 0.513)	(316, 0.533)	(311, 0.533)
$N = 500$	(196, 0.452)	(225, 0.463)	(229, 0.462)	(275, 0.484)	(274, 0.514)	(334, 0.536)	(333, 0.540)
$N = 1000$	(186, 0.454)	(207, 0.454)	(236, 0.468)	(271, 0.484)	(274, 0.516)	(345, 0.542)	(371, 0.551)
$N = 2000$	(207, 0.463)	(236, 0.467)	(242, 0.463)	(257, 0.492)	(295, 0.511)	(359, 0.547)	(392, 0.548)

Table 33: UDLM: the (novel, novel-QED) point at each λ , for every N . Reading a row left to right traces that curve.

D-CBG						
exact	$\gamma = 1$: (118, 0.528)	$\gamma = 2$: (115, 0.567)	$\gamma = 3$: (102, 0.588)	$\gamma = 5$: (82, 0.593)	$\gamma = 10$: (63, 0.609)	
approx.	$\gamma = 1$: (121, 0.477)	$\gamma = 2$: (116, 0.485)	$\gamma = 3$: (95, 0.523)	$\gamma = 5$: (78, 0.529)	$\gamma = 10$: (59, 0.556)	

Table 34: UDLM: the same for D-CBG, sweeping γ . The novel count falls as the property rises — the opposite direction.

10 What the sweep says

MDLM. Peak novel-QED: ours 0.520 against D-CBG 0.588. Peak novel count: ours 439 against D-CBG 208. Each method wins one axis, so the comparison is a genuine frontier. **UDLM.** Peak novel-QED: ours 0.556 against D-CBG 0.609. Peak novel count: ours 397 against D-CBG 121. Each method wins one axis, so the comparison is a genuine frontier.

10.1 Where the gains come from

λ and N are not interchangeable. λ sets *how hard* the tilt pulls and saturates once the weights concentrate; N sets *how good the best available candidate is*, and along the high- λ edge the property still rises with N at $N = 500$. The two together trace the frontier: high λ with small N is a greedy sampler with nothing good to pick from, while large N with $\lambda = 0$ is just the base model.